Roll No. | | | | | | | |

END SEMESTER EXAMINATION JUNE – 2025

Name of the Course: B.Tech Semester: II

Name of the Paper: Programming for Problem Solving

Paper Code: **TCS201**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) Q1. is having only two parts. Attempt both the parts.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Each question carries 20 marks.

Q.1 **(20 Marks)**

a) *Assuming that the given segments of C and python programs does not prompt any compile time and/or run time errors, predict the output for them, when the following segments of code are executed on a 32-bit machine.* **[10]** CO1 CO2 CO3 CO4 CO5 CO6

1.
```c
#include <stdio.h>
int main()
{
  int p[5]={1,3,2},i,t,*a;
  a=&p[0];
  for(i=0;i<4;i++)
  {
     if(*(a+i)>*(ai+1))
     {
        t=*(a+i);
        *(a+i)=*(a+i+1);
        *(a+i+1)=t;
        i=-1;
     }
  }
  for(i=0;i<=4;i++) printf("%d",a[i]);
   return 0;
}
```

2.
```c
#include <stdio.h>
#include<string.h>
int main()
{
   char *s="GEU";
   char s1[10]="GEU";
   if(s==s1)
      printf("1. s == s1\n");
   else
      printf("1. s != s1\n");
   if(!strcmp(s,s1))
      printf "2. s == s1\n");
   else
      printf("2. s != s1\n");
   return 0;
}
```

3.
```c
#include <stdio.h>
int main()
{
  int a[3][3]={0,1,1,1,0,1,1,1,0},i,j;
  for(i=0;i<3;i++)
  {
     for(j=0;j<3;j++)
     {
        a[i][j]=!a[i][j];
     }
  }
  printf("%d", a[0][0]+a[1][1]+a[2][2]);
  return 0;
}
```

4.
```c
#include <stdio.h>
typedef struct date
{
   unsigned int dd:5;
   unsigned int mm:4;
   unsigned int yy:11;
}date;
int main()
{
   date d1={31,12,2024};
   date d2={20,20,20};
   printf("Sizeof date is: %d\n",sizeof(struct date));
   printf("d1 is: %d:%d:%d\n",d1.dd,d1.mm,d1.yy);
   printf("d2 is: %d:%d:%d\n",d2.dd,d2.mm,d2.yy);
   printf("%d", sizeof(d1));

}
```

5.
```c
#include<stdio.h>
#include<string.h>
int main()
{
int i,t,a[5]={5,4,3,2,1};
FILE *fp=fopen("1.txt","w");
for(i=0;i<5;i++)
{
   fprintf(fp,"%d\n", a[i]);
    fprintf(stdout,"%d\t", a[i]);
}
printf("\nBye");
}
```

6.
```c
#include<stdio.h>
int main()
{
  int x = 20, *y, *z;
  y = &x;
  z = y;
  (*y)++;
  --(*z);
  x++;
  printf("x = %d, y = %d, z = %d \n", x, *y, *z);
  return 0;
}
```

7.
```c
#include<stdio.h>
#include<string.h>
int main()
{
char s[20]="Graphic Era";
int l,i,j,t;
l=strlen(s);
for(i=0,j=l; i<j; i++,j--)
{
        t=s[i];
        s[i]=s[j];
        s[j]=t;
}
printf("The string is: %s\n",s);
}
```

8.
```c
#include <stdio.h>
int main()
{
  float A[3][3]={1.1,-10,2.3, 6.1, 0.003, 4.62, 7.54};
  float sum=0;
  unsigned static int i,j=2;
  do
  {
     sum + = A[i++] [j--];
  } while(i<j);
  printf("%.2f, %d\n", sum, (int)sum);
  return 0;
}
```

9.
```python
l = ["Graphic", "Era", "Dehra Dun"]
for i in l:
 print(i)
 print('i')
```

10.
```python
l = list (range (1,100))
i = 1
length=(len(l))
print(length)
while i < length:
   print(i)
   i = i + i
```

b) Draw a flowchart to read a string str and store in the memory allocated at run time using DMA. Pass this string into a function named *Modify()* which modify string by replacing all the special characters with the character '#'. **[10]**

Q.2 **(20 Marks)**

a) Write a 'C' program to input two matrix of size *mXn* and check whether second matrix is row-wise mirror image of first matrix. If yes then display "Both matrices are same" otherwise display "Matrices are not same". **[10]** CO1 CO3

Sample Input:	Sample Output:
Row size: 2 Col size: 3 Elements of 1st matrix: Elements of 2nd matrix: 1 2 3 3 2 1 4 5 6 6 5 4	Both matrices are same

b) State the purpose of Dynamic Memory Allocation (DMA) in 'C'. What are the various methods to achieve DMA? Explain with suitable syntax and example. Which header file must be included to use these functions? When does a memory leak arise? What precaution must a programmer practice to avoid it? **[10]**

c) Develop a 'C' program to read *n* integer numbers and store in the array created using dynamic memory allocation. Check each integer number stored in the array. If number is even then replace it with its next even number and if the number is odd then replace it with its previous odd number. Print the final array. [10]

Sample Input:	Sample Output:
Value of *n*: 5 Elements of array: 2 5 1 8 9	Final array after changes: 4 3 -1 10 7

Q.3 **(20 Marks)**

a) Describe structure? What is the purpose of dot and arrow operator in structure. Explain with suitable example. Also describe the following terms with suitable example. [10]

i) self-referential structure ii) bit-field iii) structure within structure

b) Management of an organization wants to fetch the detail of working hours of their employee on a particular day. Help management team with a 'C' code which will input detail of *n* employees of the organization. Display their complete detail and also find working hour of each employee.

Note: Create a structure *Employee* with the data members *Emp_name(char type)*, *Emp_id(int type)*, *PunchInTime(int)*, *PunchOutTime(int)*. (consider the format: *hr min sec* for PunchInTime and PunchOutTime). [10] CO4

c) Create a structure *City* having data members *City_name(char type)*, *Male_count(int type)* and *Female_count(int type)*. Write a 'C' code to read detail of 15 cities. Display detail of all the city along with total population. Also print difference between gender count of each city with displaying a message "*Male count more*" or "*Female count more*". [10]

Q.4 **(20 Marks)**

a) Explain the need of file in 'C' programming. Explain the following methods with their suitable syntax and purpose. [10]

i) fopen()
ii) fwrite()
iii) fseek()
iv) fgets()

b) A small factory wants to maintain the record of its workers month wise in a file named "DATA". Record contains *Worker_Name(char type)*, *Worker_Id(int type)*, *Year_of_Exp(int type)*, *Pay_per_Day(int type)* and *No_of_days(int type)*. Write a 'C' program to write detail of 25 workers in the file. Extract the content of the file "DATA" and display the detail of all the workers along with their *Gross_Salary* and *Net_Salary*. [10] CO5

Gross salary = pay * days

Net salary = Gross salary – epf

If year of experience >10 then epf = 2000

If year of experience >5 then epf = 1500

If year of experience >1 then epf = 1000

Otherwise epf = 0

c) Develop a 'C' code to write *n* integer numbers into a file named "DATA_1". Read the file and copy the all the perfect square numbers to another file named "DATA_2". [10]
[10]

Sample Input:	Sample Output:
Content of DATA_1: 2 3 4 7 9 14 16 18 20	Content of DATA_2: 4 9 16

Q.5 **(20 Marks)**

a) "Python is a popular programming language". Justify your answer. How it is different from 'C' language? Explain various data types available in python with proper example. Also explain any two categories of operators available in Python. [10]

b) Develop a menu driven Python code which use multiple functions for the following menus: CO6

1) *func1()* accept a character and print a message "It is an Alphabet" and "It is not an Alphabet"
2) *func2()* accept a four-digit integer number and replace its first and last digit. [10]

c) Develop a python code which display all the palindrome prime numbers in the range p and q.

Note: Palindrome prime number are the numbers which are prime as well as palindrome also.

Example: 2 5 7 11 101 131 [10]